SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3 **Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I **C.BHUVANESH (192521108)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: c.bhuvanesh

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

1. Internet Connection

Primary: 4G/5G Fixed Wireless Access (FWA)

- Use a business-grade 4G/5G router with an external high-gain antenna.
- Provides sufficient bandwidth for cloud applications, email, VoIP, and video conferencing.
- Faster and cheaper to deploy than laying fiber or leased lines.
- Can be upgraded from 4G to 5G as coverage improves.

Backup: Satellite Internet

- Use satellite Internet as a failover connection when the cellular network becomes unavailable.
- Provides connectivity even in locations with poor terrestrial infrastructure.
- Although satellite has higher latency and installation costs, it improves business continuity.

Recommended approach:

```
4G/5G → Primary
Satellite → Backup
```

2. Networking Devices

Device	Purpose
4G/5G Router/Modem	Connects the office to the cellular network
Satellite modem	Provides backup Internet
Dual-WAN Router	Automatically switches between primary and backup connections
Managed Switch	Connects PCs, printers, IP phones, and servers
Wi-Fi 6/6E Access Points	Provides wireless connectivity to employees
Firewall/UTM	Protects the network from Internet threats
VPN Gateway	Creates a secure connection to headquarters
UPS	Keeps network devices running during short power failures

3. Security Mechanisms

Firewall

- Blocks unauthorized incoming traffic.
- Controls outgoing traffic and prevents unwanted applications from consuming bandwidth.

Site-to-Site VPN

- Establishes an encrypted tunnel between the rural branch and headquarters.
- Protects company data from interception over cellular or satellite networks.

WPA3 Wi-Fi Security

- Use WPA3 for secure wireless access.
- Create separate employee and guest Wi-Fi networks.

Network Segmentation

- Separate business computers, IP phones, IoT devices, and guest users using **VLANs**.
- This limits the impact if one device is compromised.

MFA and Access Control

- Use multi-factor authentication for cloud applications and administrative accounts.
- Give employees only the permissions they need.

Monitoring and Updates

- Regularly update router, firewall, and access-point firmware.
- Monitor bandwidth, connection failures, and security events.

4. Reliability

Since uninterrupted operations are required, the network should use **redundancy**:

```
            ┌── 4G/5G Internet ──┐
Office LAN ── Firewall/Dual-WAN Router ── Internet
            └── Satellite ──────────┘

                 ↑

             Automatic

              Failover
```

If the 4G/5G connection fails, the dual-WAN router automatically moves traffic to the satellite connection.

A **UPS** should also protect the router, firewall, switch, and Wi-Fi equipment from short power interruptions.

5. Bandwidth Management

Because wireless connections may have limited bandwidth:

- Give **video conferencing and business-critical cloud applications** higher QoS priority.
- Give normal web browsing and downloads lower priority.
- Use traffic monitoring to identify bandwidth-heavy applications.
- Consider bandwidth limits for guest Wi-Fi.
- Use caching and cloud optimization where appropriate.

For example:

Application	**Priority**
Video conferencing / VoIP	**High**
Cloud business applications	**High**
Email	Medium
Web browsing	Medium
Large downloads / updates	Low

6. Cost Considerations

A dedicated fiber or leased line would provide excellent performance but may be **expensive or unavailable** in a rural location.

The recommended solution has a lower initial deployment cost:

4G/5G + business router + firewall + managed switch + Wi-Fi APs

Satellite is added only as a backup, so its bandwidth and subscription cost can be kept under control.

7. Future Scalability

The design can easily be expanded by:

- Upgrading from 4G to 5G.
- Adding additional Wi-Fi access points.
- Adding more switches and VLANs.
- Increasing Internet bandwidth.
- Adding more VPN connections to headquarters.
- Supporting additional cloud services, IP phones, and IoT devices.
- Adding another WAN connection if business requirements increase.

Conclusion

The best solution is a **4G/5G fixed wireless connection as the primary link with satellite as a backup**, managed by a **dual-WAN router and enterprise firewall**. A managed switch, Wi-Fi 6 access points, VPN, VLANs, WPA3, QoS, and UPS provide security, performance, and reliability.

This solution offers a good balance of **bandwidth, reliability, cost, security, and future scalability** without requiring expensive wired infrastructure.

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design,

including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

1. Access Point Placement

Access points should be placed according to **user density and building structure**.

- **Classrooms:** Place APs near areas with high student density. Avoid placing one AP to serve several crowded classrooms through thick walls.
- **Laboratories:** Install APs where many computers and wireless devices are used.
- **Libraries:** Use APs with careful channel planning because many students may simultaneously use laptops and phones.
- **Hostels:** Deploy APs by floor/wing based on the expected number of users.
- **Common areas:** Place APs in locations such as cafeterias and study areas where large numbers of students gather.
- Perform a **wireless site survey** before installation to identify dead zones, walls, interference, and signal strength.

Instead of deploying APs uniformly, give more AP capacity to **high-density areas**.

2. Frequency Band Selection

Use **dual-band or tri-band Wi-Fi 6/6E access points**.

Band	Use	Advantage
2.4 GHz	Basic connectivity and older devices	Longer range
5 GHz	Main student traffic	Higher speed and more channels
6 GHz	Compatible modern devices	More clean spectrum and less congestion

5 GHz should be the primary band for most students because it provides better performance and more usable channels than 2.4 GHz.

The **6 GHz band** can be used for newer devices in high-density areas such as libraries and laboratories.

2.4 GHz should be used selectively because it has fewer non-overlapping channels and is more susceptible to interference.

3. Channel Planning and Interference Reduction

To minimize interference:

- Use **automatic or centrally managed channel assignment**.
- Avoid using the same channel on nearby APs where possible.
- Reduce excessive AP transmit power so neighboring APs do not interfere with each other.
- Prefer **5 GHz/6 GHz** for high-density users.
- Use **band steering** to encourage compatible devices away from congested 2.4 GHz.
- Use **Wi-Fi 6 features such as OFDMA** to efficiently serve many clients.
- Monitor the wireless environment continuously and change channels when interference increases.

4. Authentication and Security

The university should use **WPA3-Enterprise with 802.1X authentication**.

A typical authentication process is:

Student Device

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Wi-Fi Access Point

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802.1X / RADIUS Server

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University Identity Database

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Authenticated Network Access

Students can authenticate using their **university username/password or institutional credentials**.

Security should include:

- **WPA3-Enterprise** encryption
- **802.1X authentication**
- **RADIUS server** for centralized authentication
- Separate **student, staff, guest, and IoT VLANs**
- Firewall policies between VLANs
- Guest network with Internet-only access
- Regular monitoring and firmware updates

5. Network Architecture

A centralized architecture would be efficient:

```
        Internet
           |
      Core Network
           |
  Wireless LAN Controller
     /     |     \
    AP     AP     AP
   /|\    /|\    /|\
```

Classrooms Library Hostels

A **Wireless LAN Controller (WLC)** or cloud-managed WLAN platform can centrally manage APs, channels, transmit power, security, and roaming.

6. Handling 3,000+ Students

The important point is that APs should be selected based on **capacity**, not just coverage.

For example, if an AP can practically support around 50–75 active clients under the university's required performance level, the network might require roughly **40–60 APs** for 3,000 students, depending heavily on simultaneous usage and building conditions.

The exact number should be determined through a **site survey and capacity analysis** rather than simply dividing the student population by an AP's theoretical maximum.

7. Balancing Coverage, Performance, Security, and Cost

Requirement	Proposed Solution
Coverage	Strategic AP placement and site survey
Large number of users	High-capacity Wi-Fi 6 APs
Performance	5 GHz/6 GHz + OFDMA
Interference	Channel planning and transmit-power control
Security	WPA3-Enterprise + 802.1X

Authentication	RADIUS + university credentials
Cost	Deploy APs mainly where user density is high
Scalability	Centralized WLAN management
Guest access	Separate guest SSID/VLAN

Conclusion

The most efficient design is **not to install the maximum possible number of APs**. Instead, the university should use a **site-survey-based, high-density Wi-Fi 6 design**, concentrating APs in classrooms, laboratories, libraries, and hostels where user density is highest.

Using **5 GHz as the primary band, selective 6 GHz deployment, controlled 2.4 GHz usage, careful channel planning, WPA3-Enterprise, 802.1X/RADIUS authentication, VLAN segmentation, and centralized AP management** provides a good balance of **coverage, performance, security, and cost**.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

1. Proposed Network Architecture

A suitable design is:

```
              Internet
                 |
            [Edge Router]
                 |
           [Perimeter Firewall]
                 |
      +-------------+-------------+
      |                           |
   DMZ Network            Internal Firewall
      |                           |
 Web / Online Banking        Core Network
   Servers                        |
               +--------+--------+
               |        |        |
             VLAN 10   VLAN 20   VLAN 30
             Staff     Banking   IT/Admin
               |        |        |
           Employees  Sensitive   Admin
                      Systems    Systems
                        |
               [Database Network]
                        |
               Customer Database
```

2. Network Segmentation

Use **VLANs and subnetting** to separate different types of users and systems.

Segment	Purpose	Security Level

DMZ	Public-facing online banking/web servers	High
Employee VLAN	Normal employee computers	High
Banking VLAN	Sensitive banking applications	Very High
Database VLAN	Customer and transaction databases	Critical
Management VLAN	Network/security administration	Critical
Guest VLAN	Guest Internet access	Low

Why segmentation?

If an employee's computer is compromised, the attacker should not automatically be able to reach the customer database. Access between VLANs should pass through firewall rules.

This provides strong security without requiring separate physical networks, helping control costs.

3. Firewall Deployment

Use a **next-generation firewall (NGFW)** at the Internet boundary.

The firewall should provide:

- Stateful packet inspection
- Application-level filtering
- Intrusion prevention
- VPN support
- URL/content filtering
- Logging and monitoring

A second internal firewall can protect the **sensitive banking/database network**.

For example:

Internet

↓

Perimeter Firewall

↓

DMZ

↓

Internal Firewall

↓

Employee / Banking VLANs

↓

Database Firewall Rules

↓

Customer Database

Only required traffic should be permitted.

For example:

- Employees → Banking application: **Allow**
- Employees → Database directly: **Deny**
- Internet → Database: **Deny**
- Banking application → Database: **Allow only required ports**
- Guest → Internal network: **Deny**

4. Authentication

Use **Multi-Factor Authentication (MFA)** for sensitive systems.

Recommended approach:

Something you know: Password
Something you have: Security token/mobile authenticator
Optional: Biometric authentication

For internal network access, use **802.1X authentication** with a centralized **RADIUS/AAA server**.

This ensures that only authorized employees can connect to protected network resources.

Also implement:

- Role-Based Access Control (RBAC)
- Strong password policies
- Account lockout policies
- Least-privilege access
- Regular review of employee permissions

5. Intrusion Detection and Prevention

Deploy both **IDS and IPS**.

IDS (Intrusion Detection System):

- Monitors network traffic.
- Detects suspicious activity.
- Generates alerts for security administrators.

IPS (Intrusion Prevention System):

- Detects malicious traffic.
- Can automatically block or drop suspicious packets.

A practical deployment is:

Internet

↓

Firewall / IPS

↓

DMZ

↓

Internal Network

↓

IDS Sensors

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SIEM / Security Monitoring

A **SIEM** can collect firewall, IDS/IPS, authentication, and server logs in one place for security monitoring and compliance reporting.

6. Performance Considerations

Security should not significantly slow down banking operations.

To maintain performance:

- Use hardware-accelerated firewalls where appropriate.
- Avoid unnecessary deep inspection on trusted, low-risk traffic.
- Apply security inspection selectively to sensitive networks.
- Use VLANs to reduce broadcast traffic.
- Monitor bandwidth and firewall CPU/memory utilization.
- Use redundant security devices for critical services.

7. Compliance

Financial organizations need strong controls around sensitive information and access.

The architecture supports compliance through:

- Encryption of sensitive communications.
- MFA for privileged and sensitive access.
- Centralized authentication.
- Detailed security logging.
- Regular vulnerability scanning.
- Access-control policies.
- Network segmentation.
- Audit trails through SIEM.
- Regular backup and recovery testing.

The exact regulatory requirements depend on the institution's country and applicable standards, so the controls should be mapped to the organization's specific compliance framework.

8. Budget Considerations

The organization wants better security without significantly increasing operational costs.

Therefore:

- Use **VLANs** instead of physically separate networks wherever possible.
- Use one centrally managed firewall platform rather than many individual firewalls.
- Use existing switches that support VLANs and 802.1X.
- Centralize authentication using RADIUS/AAA.
- Use open-source or existing SIEM/IDS capabilities where appropriate, provided they meet compliance and support requirements.
- Automate monitoring and security alerts to reduce administrative workload.

9. Recommended Security Layers

Layer	Security Mechanism
Internet	Edge firewall
Public services	DMZ
Internal network	VLAN segmentation
Sensitive systems	Internal firewall

User access	802.1X + RADIUS
Sensitive applications	MFA + RBAC
Network threats	IDS/IPS
Monitoring	SIEM
Data	Encryption
Administration	Separate management VLAN

Conclusion

The recommended architecture uses **VLAN-based segmentation, a perimeter NGFW, an internal firewall for critical systems, 802.1X/RADIUS authentication, MFA, and IDS/IPS with centralized SIEM monitoring**.

This provides **strong defense-in-depth**: segmentation limits the spread of attacks, firewalls control communication, MFA prevents unauthorized account use, and IDS/IPS detects and blocks threats.

At the same time, using logical segmentation, centralized management, existing network infrastructure, and selective security inspection keeps **performance and operational costs under control** while supporting the financial institution's security and compliance requirements.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and	Identifies most constraints with minor	Identifies some constraints but	Shows little or no understanding

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
		fully understands the problem requirements	gaps in understanding	misses key aspects	of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand